

Dipu Kumar Sharma

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PROFESSIONAL SUMMARY

Dedicated Python and Node.js developer with over 5 years of experience specializing in web development, backend systems, and team leadership. Expertise in Fastapi, Django, Generative AI, and Retrieval-Augmented Generation (RAG) architectures to deliver efficient, scalable solutions. Proven track record of leading cross-functional teams, managing database architectures, and driving innovation in software delivery.

TECHNICAL SKILLS

- **Backend Frameworks:** FastAPI, Django, Express.js, Nest.js, Nuxt3, Chalice
- **Programming Languages:** Python, JavaScript, TypeScript, Go, XML
- **AI & Data:** Generative AI, Agent AI, Agentic AI, LLMs, RAG, Ollama, Chroma DB, Qdrant, Machine Learning, Data Science, Data Analysis, LangChain, LangGraph
- **Databases:** PostgreSQL, MongoDB, SQL, MySQL, DynamoDB, Supabase, Redis
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, CloudWatch), GCP, Azure, Docker, Celery, RabbitMQ
- **Frontend & Other:** Vue.js, React.js, Tailwind CSS, Odoo, Web Scraping, Git

WORK EXPERIENCE

Software Developer | DEVtrust *June 2025 – Present*

- **E-Abhilekh:** Engineered a document management application for departmental PDF processing using Django and GCP. Integrated an AI Smart Search feature utilizing Generative AI, LLMs, and Ollama to query uploaded documents. Utilized Qdrant for vector storage and implemented automated draft generation (Notices, Circulars, Summaries) based on file context.
- **Updesh Chatbot App:** Architected a Retrieval-Augmented Generation (RAG) system using FastAPI, Nuxt.js, and Chroma DB. Developed a document parsing pipeline with Docling to extract, chunk, and embed large text files securely, ensuring the chatbot generates strictly contextual responses without hallucinating false records.
- **ATR Service:** Developed an order management system utilizing the AWS Chalice framework, AWS Lambda, DynamoDB, and S3. Integrated GraphQL mutations and queries via webhooks to manage third-party services (Qualia).
- **Edu-Tech:** Built a real-time student management and chat platform using Django 5, Redis, Celery, and WebSockets. Implemented dynamic teacher-student group messaging and secure, dedicated channel creation functionalities.
- **File-Parser:** Created a high-performance data extraction tool using Go to map CSV and PDF data to relational schemas. Utilized Supabase and PostgreSQL window/aggregation functions to calculate alias columns and prevent duplicate database records.

Web Team Lead | UniteCloud *May 2024 – May 2025*

- **RR Parkon:** Led a web development team to build a construction project management system using Express.js, Nuxt.js, and MongoDB. Engineered a dynamic milestone tracking module with automated start/end date constraints and specific resource allocation.

- **Power Proton:** Directed backend development for a generator installation tracking app utilizing FastAPI and PostgreSQL. Developed admin dashboards to visualize field data collected during employee site visits.
- **Uni-Approver:** Architected a dynamic Service-Oriented Architecture (SOA) matrix for conditional request approvals using FastAPI. Implemented automated approver fetching based on dynamic form categories and hierarchical user limits.
- **Bidwise:** Built an online auction and bidding platform for scrap material using FastAPI and Vue.js. Developed features for direct vendor-to-company negotiation and automated low-bid slot displays.
- **HRMS:** Developed a Full-Stack HRMS application to analyze employee attrition and predict leave using React.js and FastAPI. Configured dynamic Odoo database connections to facilitate secure access to analytical dashboards.

Asst. Software Engineer | Dreamztech Solution Dec 2021 – Feb 2024

- **Flux Pay:** Developed the backend for a comprehensive payment gateway system (Admin, Merchant, User modules) using FastAPI, PostgreSQL, and RabbitMQ. Engineered multi-way fund collection methods including one-way, split-way, and recurring payments.
- **Sage Goddess:** Extended the Odoo framework inventory module with custom backend features including an SKU calculator, shipment tracking, and vendor ranking. Programmed ZPL commands for automated barcode generation and printing.
- **Changed Management:** Built a URL shortening service and user management system using Nest.js and Django Rest Framework. Implemented dynamic PDF generation capabilities with graph content for advanced user reporting and analytics.

Web Developer | NIBMG June 2021 – Nov 2021

- **Report-Management:** Developed a Django-based full-stack application to extract and process DNA data from uploaded ZIP/RAR archives. Engineered the database schema to store parsed text and image data, visualizing DNA reports on a React-based frontend web portal.

Web Developer | Smart Info Pvt Ltd Nov 2019 – March 2021

- **Kreedha-Pharma:** Built an online order management system for a pharmaceutical company using Django and React.js. Automated daily, monthly, and yearly sales report generation and web scraping pipelines.

EDUCATION

- Bachelor of Engineering (Computer Science & Engineering) | Rajiv Gandhi Prodyogiki Vishwavidyalaya (Bhopal, MP) 2019
- Higher Secondary (PCM) | Uttar Pradesh Board (UP - Ballia) 2013

CERTIFICATIONS

- Data Visualization and Machine Learning — Tutorials Point
- FastAPI — Udemy
- Nest.js — Tutorials Point

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